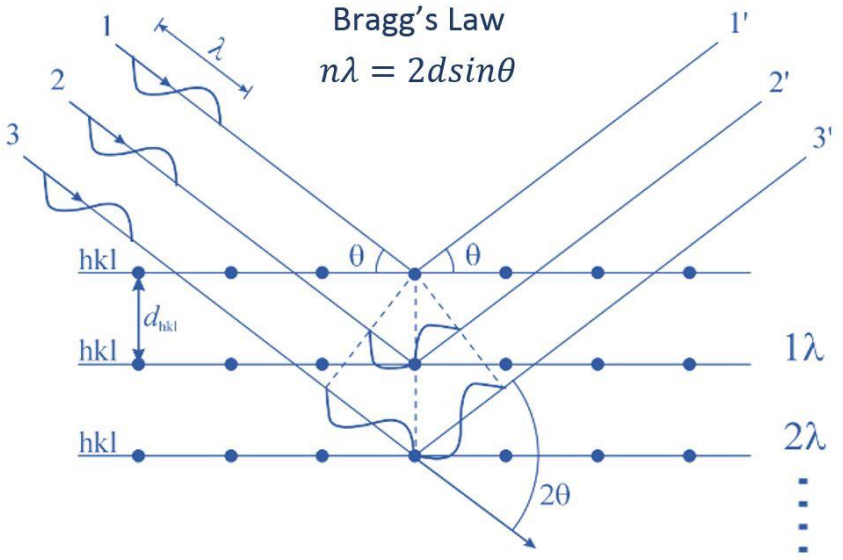
# Optics
# Chapter 3: Interference



For B1 Students
Academic year 2024-2025




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# OVERVIEW

| Chapter | Contents | Hours | | | Ref./Resources | Assignment(s) |
|---|---|---|---|---|---|---|
| | | Lect. | Exr. | Prc. | | |
| 1 | The nature and propagation of light | 2.0 | 1.0 | | | Exercises |
| 2 | Geometric optics | 5.0 | 2.0 | | | Exercises |
| 3 | Interference | 3.0 | 1.5 | | | Exercises |
| 4 | Diffraction | 3.0 | 1.5 | | | Exercises |
| 5 | Photons: Light waves behaving as particles* | | | | | Exercises |
| | | 13 | 6 | | | |

| | | |
|---|---|---|
| Assessment/ Evaluation | **Attendance/Attitude** | **10%** |
| | Exercise(s) | 0% |
| | Practicals | 0% |
| | **Mid-term test** | **30 %** |
| | **Final exam** | **60 %** |

Reference Literature:

[1] Young and Freedman – University Physics with Modern Physics 15th Edition (2020)

# Chapter 3. Interference (3.0h)

3.1 Interference and Coherent sources

3.2 Two-source interference of light

3.3 Intensity in interference pattern

3.4 Interference in thin films

3.5 The Michelson interferometer

**List of Exercises:**

In textbook “University Physics with Modern Physics” 15th Edition, Hugh D. Young, Roger A. Freedman, A. Lewis Ford (2020)

**35. 2, 3, 4, 6, 9, 11, 14, 17, 19, 26, 35**

# Chapter 3. Interference (3.0h)

## 3.1 Interference and Coherent sources

- The term **interference** refers to any situation in which two or more waves overlap in space.

**The principle of superposition:**

"When two or more waves **overlap**, the **resultant displacement** at *any point* and at *any instant* is found by **adding the instantaneous displacements** that would be *produced at the point* by the **individual waves** if each were **present alone**."



Soap bubbles in the sky in Paris

# Chapter 3. Interference (3.0h)

## 3.1 Interference and Coherent sources

### 3.1.1 Interference in two or three dimensions

- Light waves can (and do) travel in *two* or *three* dimensions, as can any kind of wave that propagates in a two- or three-dimensional medium.

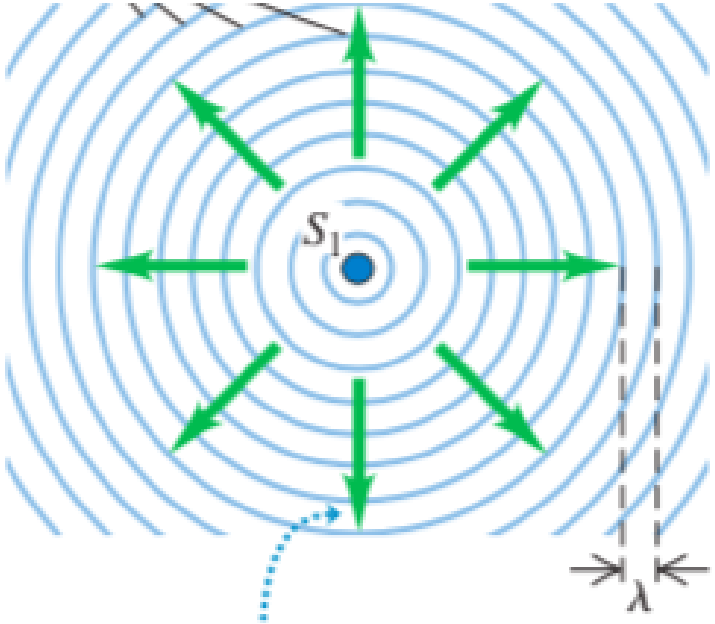
- A "snapshot" of a *single* source $S_1$ of sinusoidal waves and some of the wave fronts produced by this source:

✓ If the waves are two-dimensional,
=> the circles represent circular wave fronts;

✓ If the waves propagate in three dimensions,
=> the circles represent spherical wave fronts spreading away from $S_1$.

- In optics, sinusoidal waves are characteristic of **monochromatic light.**

Wave fronts: crests of the wave (frequency *f*) separated by one wavelength $\lambda$

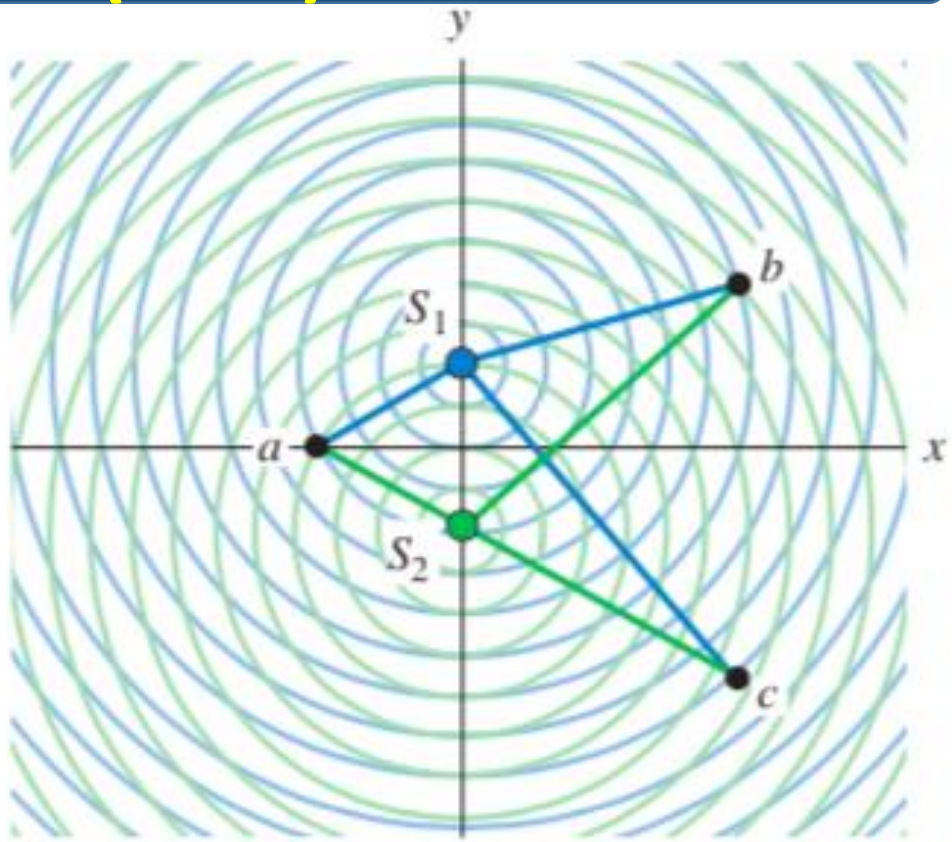



The wave fronts move outward from source $S_1$ at the wave speed $v = f\lambda$.

# Chapter 3. Interference (3.0h)

## 3.1 Interference and Coherent sources

### 3.1.2 Constructive and Destructive interference




- Two monochromatic sources of the same frequency and with a constant phase relationship are said to be **coherent.**
- The *coherent waves* (or, for light waves, *coherent light*) emitted by coherent sources.
- The two sources of equal amplitude, equal wavelength, and (if the waves are transverse) the same polarization along the *y*-axis, are positioned equidistant from the origin.
- Consider a point *a* on the *x*-axis: (or any point on *x*-axis):

The waves that leave $S_1$ and $S_2$ in phase arrive at *a* in phase.

=> the total amplitude at *a* is twice the amplitude of each individual wave.

- Consider a point *b*:

✓ The distance from $S_2$ is exactly two wavelengths greater than the distance from $S_1$ to *b*.

✓ A wave crest from $S_1$ arrives at *b* is exactly two cycles earlier than a crest emitted at the same time from $S_2$

=> the total amplitude at *b* is twice the amplitude of each individual wave.

- **Constructive interference:** when waves from two or more sources arrive at a point *in phase,* they reinforce each other: The amplitude of the resultant wave is the *sum* of the amplitudes of the individual waves.

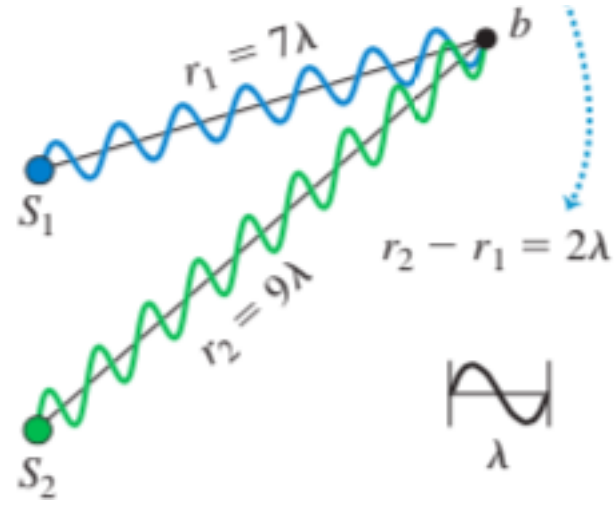
# Chapter 3. Interference (3.0h)

## 3.1 Interference and Coherent sources

### 3.1.2 Constructive and Destructive interference

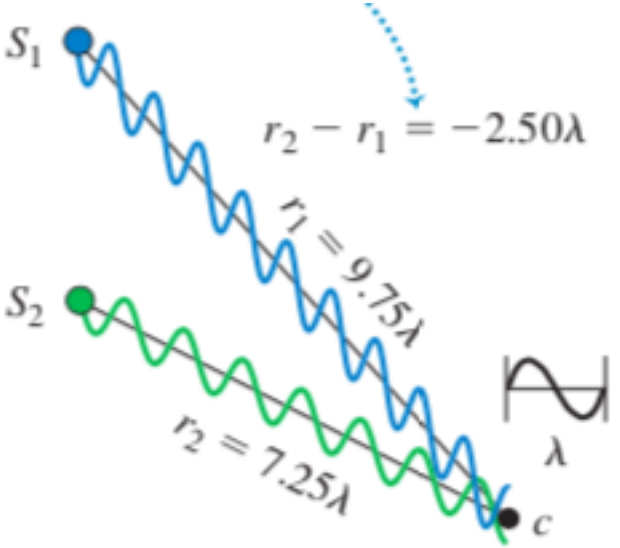
$r_2 - r_1 = m\lambda$ ($m$ = 0, ±1, ±2, ±3,…) **Constructive interference**, sources in phase

$r_1$, $r_2$: distance from $S_1$, $S_2$ to any point $P$




"For **constructive interference** to occur at $P$, the path difference $r_2$ - $r_1$ for the two sources must be an integral multiple of the wavelength $\lambda$"

$$r_2 - r_1 = \left(m + \frac{1}{2}\right)\lambda$$ ($m$ = 0, ±1, ±2, ±3,…) **Destructive interference**, sources in phase




"For **destructive interference** to occur at $P$, the path difference $r_2$ - $r_1$ for the two sources must be an half-integral multiple of the wavelength $\lambda$"

# Chapter 3. Interference (3.0h)

## 3.1 Interference and Coherent sources

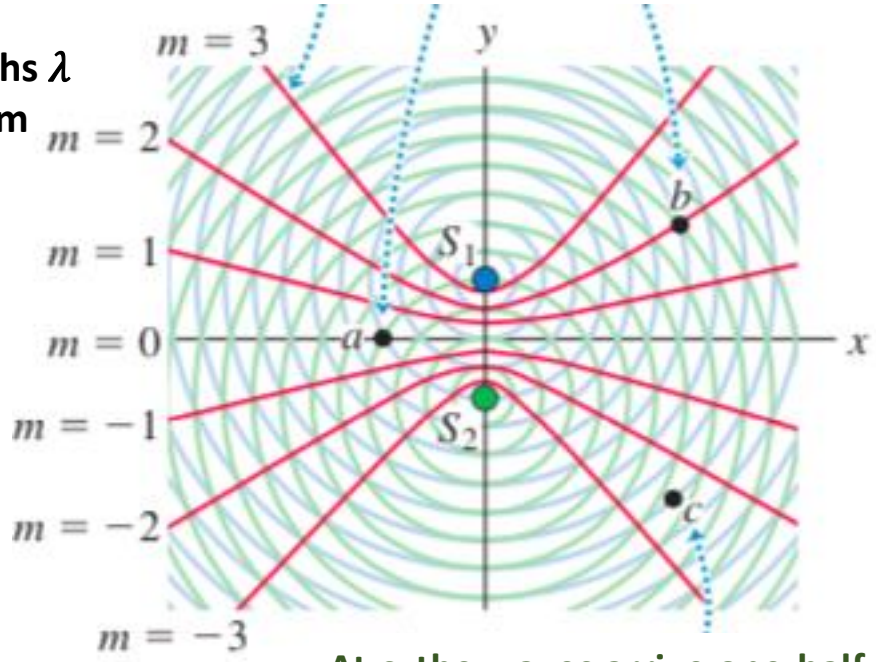
### 3.1.2 Constructive and Destructive interference

- **Red curves** that show all positions where *constructive* interference occurs: **antinodal curves**
  - On each curve, the path difference $r_2 - r_1$ is equal to an integer $m$ times the wavelength
- The **nodal curves**, which are the curves that show where *destructive* interference occurs.

**Antinodal curves (red) mark positions where the waves from $S_1$ and $S_2$ interfere constructively**

**At *a* and *b*, the waves arrive in phase and interfere constructively**

**$m$ = the number of wavelengths $\lambda$ by which the path lengths from $S_1$ and $S_2$ differ.**



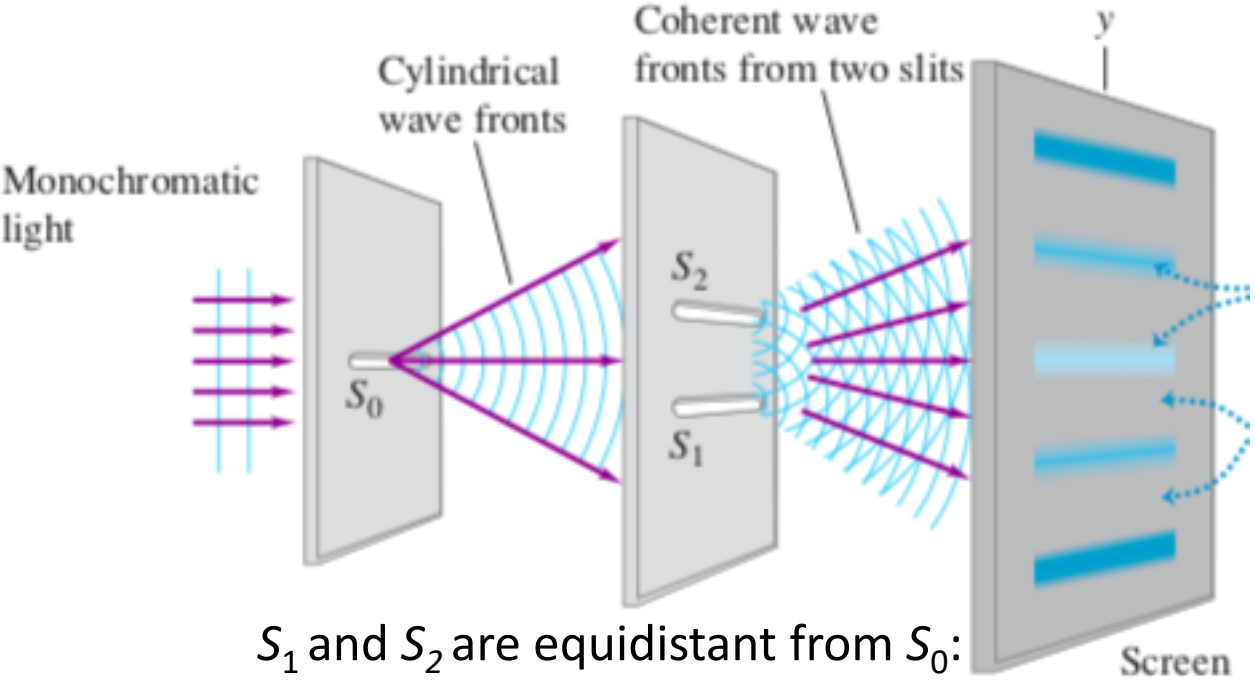

**At *c*, the waves arrive one-half cycle out of phase and interfere destructively.**

# Chapter 3. Interference (3.0h)

## 3.2 Two-source interference of light

**Young's experiment**

- 1800




$S_1$ and $S_2$ are equidistant from $S_0$: **coherent** sources

Bright bands,
Wave fronts arrive **in phase** and interfere **constructively**

Dark bands,
Wave fronts arrive **out of phase** and interfere **destructively**



Constructive and destructive interference water waves.

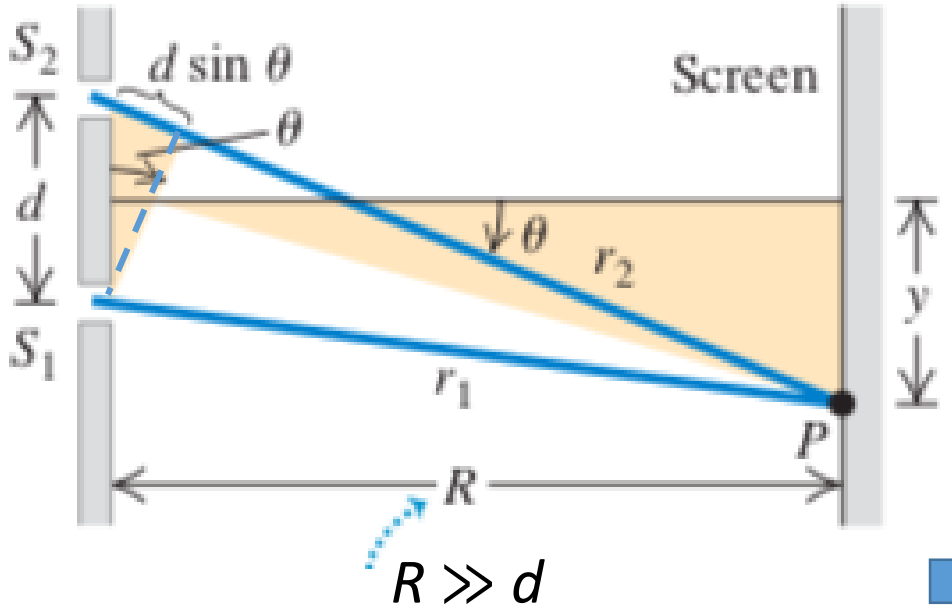
- A light source emits monochromatic light
- The light is directed at a screen with a narrow slit $S_0$ (~ 1 $\mu$m wide)
- Cylindrical wave fronts spread out from slit $S_0$ and reach $S_1$ and $S_2$ **in phase**
- The interference of waves from $S_1$ and $S_2$ produces a pattern in space at the screen.

# Chapter 3. Interference (3.0h)

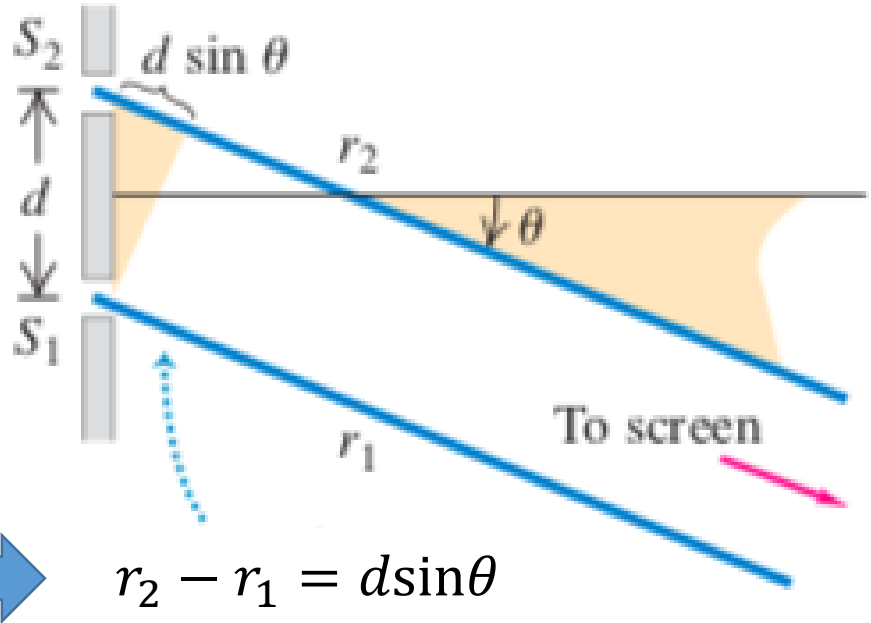
## 3.2 Two-source interference of light

### Young's experiment

- $d$: distance between two slits
- $R$: distance from the slits to the screen
- $r_1$ and $r_2$: path length from $S_1$ and $S_2$ to $P$.




$R \gg d$




$r_2 - r_1 = d\sin\theta$

- Consider $P$ – a bright point at the position $y$ on the screen
- Consider $R \gg d$ then

$$r_2 - r_1 = d\sin\theta$$

- $\theta$: the angle between a **line from slits to screen** and the **normal to the plane of the slits**.

# Chapter 3. Interference (3.

## 3.2 Two-source interference of light

### Constructive and Destructive two-slit interference

#### Constructive interference, two slits, in phase

$$d\sin\theta = m\lambda$$ ($m$ = 0, ±1, ±2,…)

$y_m$: the distance from the center of the pattern to the center of the $m$th bright band:

$$y_m = R\tan\theta_m$$

$$y_m \ll R \Rightarrow \tan\theta_m \approx \sin\theta_m$$

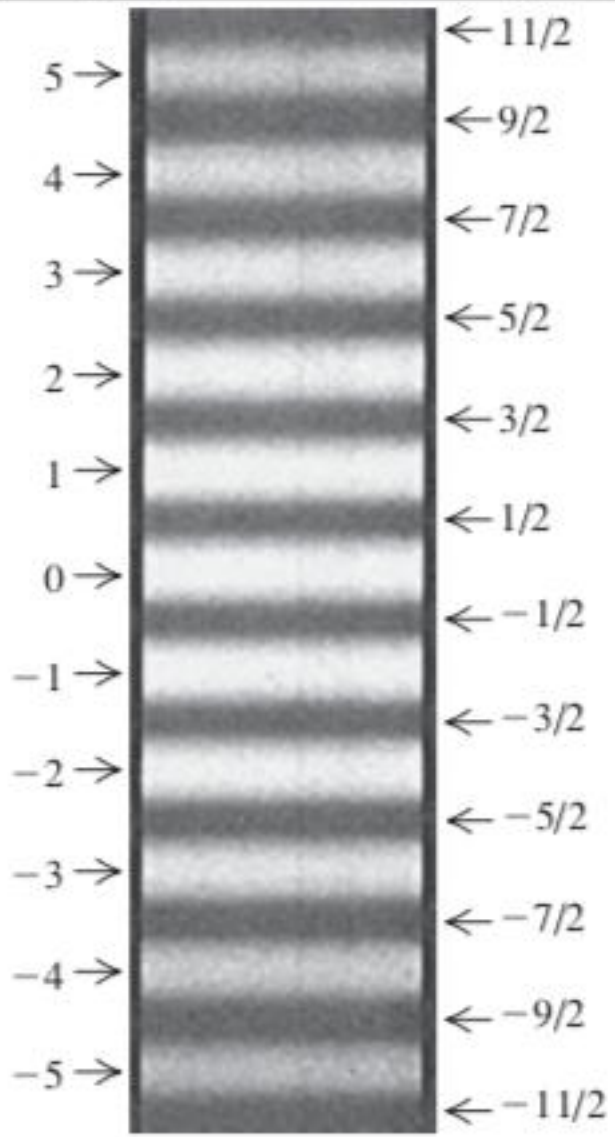
$$y_m = R\sin\theta_m$$

$$y_m = R\frac{m\lambda}{d}$$ (Small angles only)

#### Destructive interference, two slits, in phase

$$d\sin\theta = \left(m + \frac{1}{2}\right)\lambda$$ ($m$ = 0, ±1, ±2,…)

| $m$ (constructive interference, bright regions) | $m$ + 1/2 (destructive interference. dark regions) |
|---|---|




Interference fringes

# Chapter 3. Interference (3.0h)

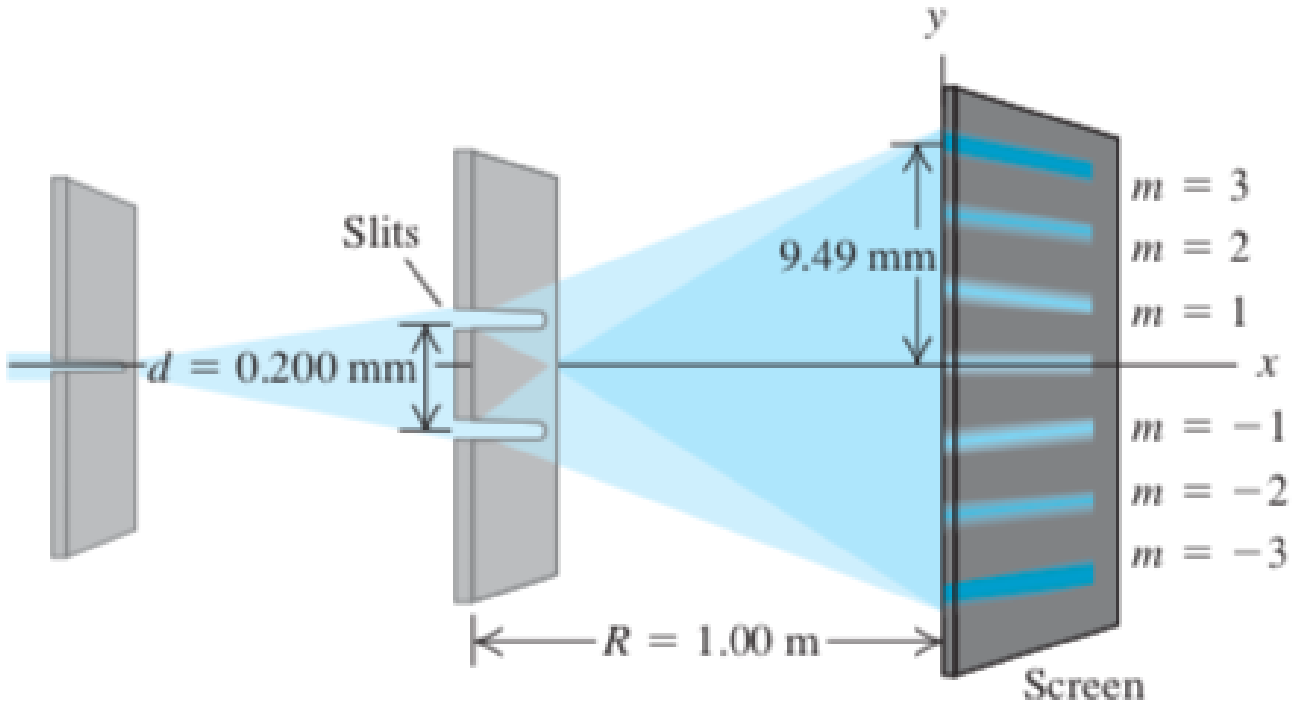
## 3.2 Two-source interference of light

**Exp.3.1 Two-slit interference**

**The figure shows a two-slit interference experiment in which the slits are 0.200 mm apart and the screen is 1.00 m from the slits. The *m* = 3 bright fringe is 9.49 mm from the central fringe.**

**Find the wavelength of the light?**

$$\lambda = \frac{y_m d}{mR} = \frac{9.49 \times 10^{-3} \times 0.200 \times 10^{-3}}{3 \times 1.00}$$
$$= 633 \times 10^{-9}\ (\mathrm{m}) = 633\ (\mathrm{nm})$$



# Chapter 3. Interference (3.0h)

## 3.2 Two-source interference of light

**Exp.3.2 Broadcast pattern of a radio station**

**It is often desirable to radiate most of the energy from a radio transmitter in particular directions rather than uniformly in all directions. Pairs or rows of antennas are often used to produce the desired radiation pattern. As an example, consider two identical vertical antennas 400 m apart, operating at 1500 kHz (near the top end of the AM broadcast band) and oscillating in phase. At distances much greater than 400 m, in what directions is the intensity from the two antennas greatest?**

- The wavelength is

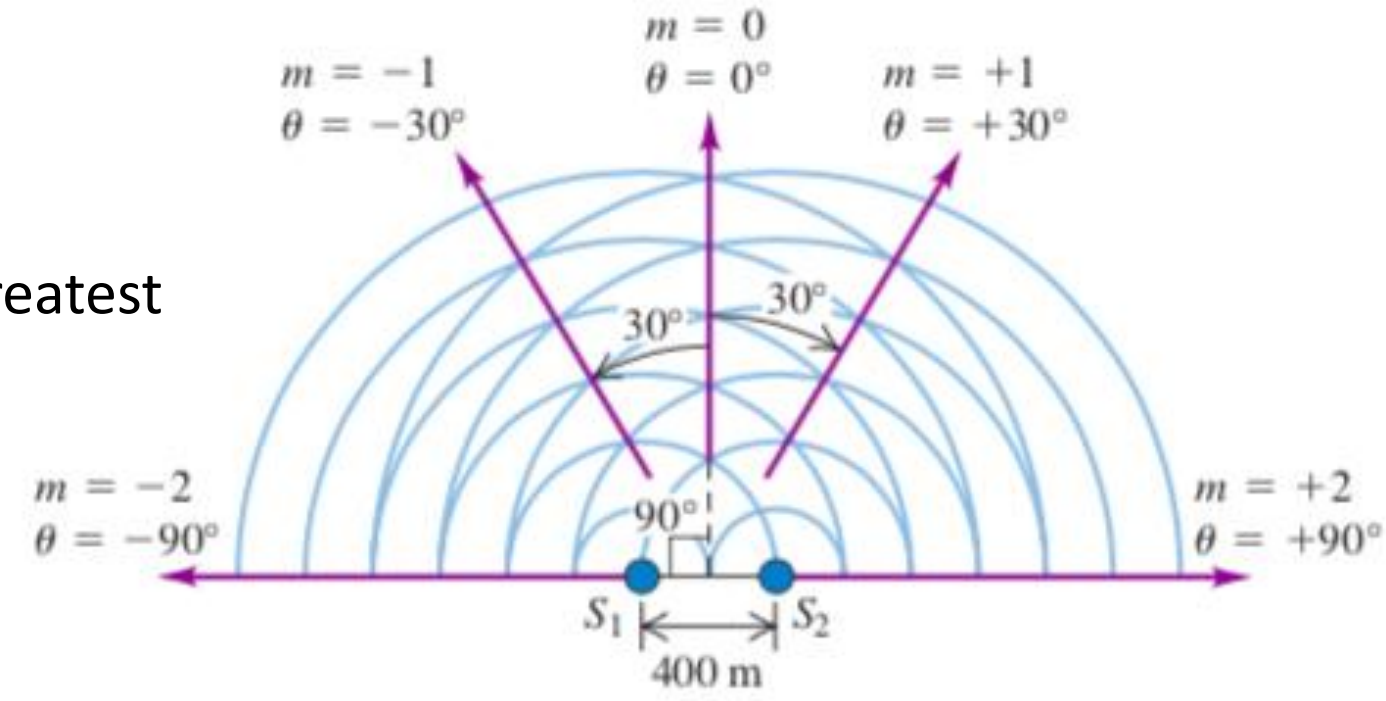
$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{1500 \times 10^3} = 200(\text{m})$$

- The intensity from the two antennas is greatest

$$d\sin\theta = m\lambda$$

$$\sin\theta = \frac{m\lambda}{d} = \frac{m \times 200}{400} = \frac{m}{2} \leq 1$$

$m = 0, \pm 1, \pm 2$ $\quad\quad$ $\theta = 0, \pm 30°, \pm 90°$



# Chapter 3. Interference (3.0h)

## 3.3 Intensity in interference patterns

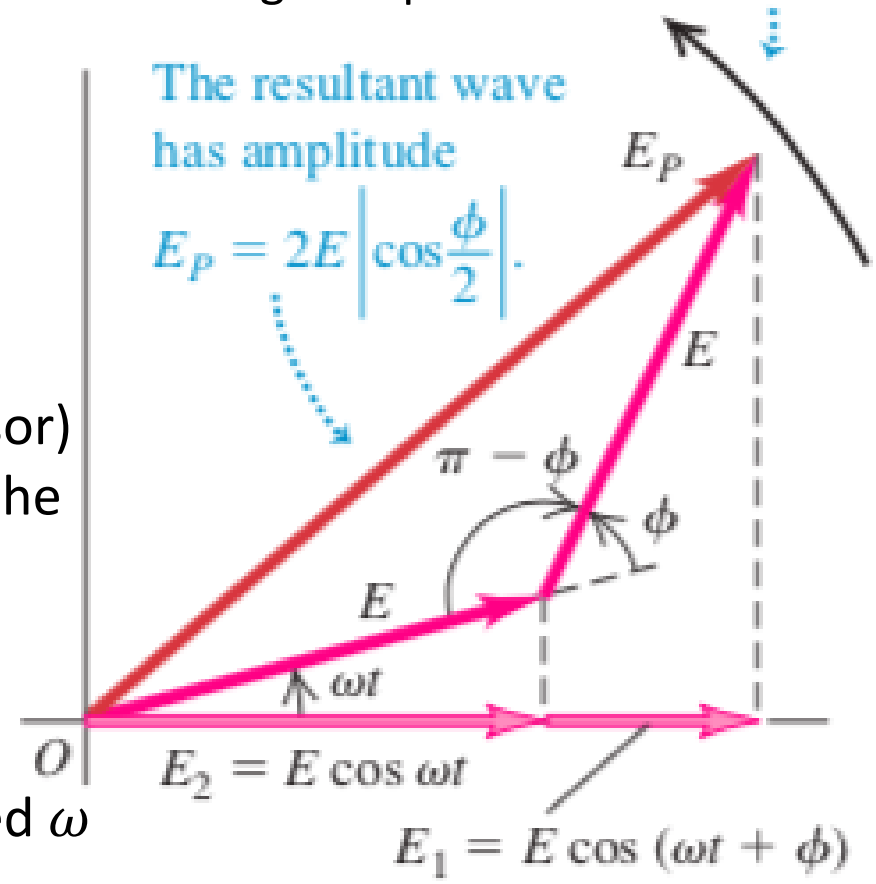
- Consider two sinusoidal functions have equal amplitude E and **E** lie along the same line (same polarization)

$$E_1(t) = E\cos(\omega t + \phi)$$
$$E_2(t) = E\cos(\omega t)$$

**Amplitude in two-source interference**

- "Each sinusoidal function is represented by a rotating vector (phasor) whose projection on the horizontal axis at any instant represents the instantaneous value of the sinusoidal function".
- Both phasors have the same magnitude *E*
- $E_1$ is ahead of $E_2$ in phase by an angle $\phi$.
- Both phasors rotate counterclockwise with constant angular speed $\omega$
- The resultant sinusoidal wave at *P*: phasor $E_P$

All phasors rotate counterclockwise with angular speed $\omega$




$$E_P^2 = E^2 + E^2 - 2E^2\cos(\pi - \phi) = E^2 + E^2 + 2E^2\cos(\phi) = 4E^2\cos^2\left(\frac{\phi}{2}\right)$$

$$\boxed{E_P = 2E\left|\cos\left(\frac{\phi}{2}\right)\right|}$$

# Chapter 3. Interference (3.0h)

## 3.3 Intensity in interference patterns

### Intensity in two-source interference

- Intensity *I* at point *P*

$$I = S_{av} = \frac{E_P^2}{2\mu_0 c} = \frac{1}{2}\sqrt{\frac{\epsilon_0}{\mu_0}}E_P^2 = \frac{1}{2}\epsilon_0 c E_P^2$$

$$I = \frac{1}{2}\epsilon_0 c E_P^2 = 2\,\epsilon_0 c E^2 \cos^2\frac{\phi}{2}$$

$\epsilon_0 = 8.854 \times 10^{-12}(\frac{\mathrm{F}}{\mathrm{m}})$: permittivity of vacuum

$\mu_0 = 4\pi \times 10^{-7}(\frac{\mathrm{H}}{\mathrm{m}})$: permeability of vacuum

- At the phase difference is zero, the maximum intensity $I_0$

$$I_0 = 2\epsilon_0 c E^2$$

- Intensity *I* at any point in terms of the maximum intensity $I_0$

$$\boxed{I = I_0 \cos^2\frac{\phi}{2}}$$

# Chapter 3. Interference (3.0h)

## 3.3 Intensity in interference patterns

### Phase difference and Path difference

- The phase difference $\phi$ between the two fields at any point *P*.

$$\phi = \frac{2\pi}{\lambda}(r_2 - r_1) = k(r_2 - r_1)$$

$k = \frac{2\pi}{\lambda}$: wave number

$r_2$ - $r_1$: path difference

- If the material between the sources and *P* not vacuum, with the index of refraction *n*

$$\lambda = \frac{\lambda_0}{n} \qquad k = nk_0$$

$\lambda_0, k_0$: the wavelength and the wave number in vacuum

- *P* is far away so that

$$\phi = k(r_2 - r_1) = kd\sin\theta = \frac{2\pi d}{\lambda}\sin\theta$$

$$I = I_0\cos^2\frac{kd\sin\theta}{2} = I_0\cos^2\left(\frac{\pi d\sin\theta}{\lambda}\right) \xrightarrow[\sin\theta_m \approx y/R]{y \ll R} I = I_0\cos^2\left(\frac{kdy}{2R}\right) = I_0\cos^2\left(\frac{\pi dy}{\lambda R}\right)$$

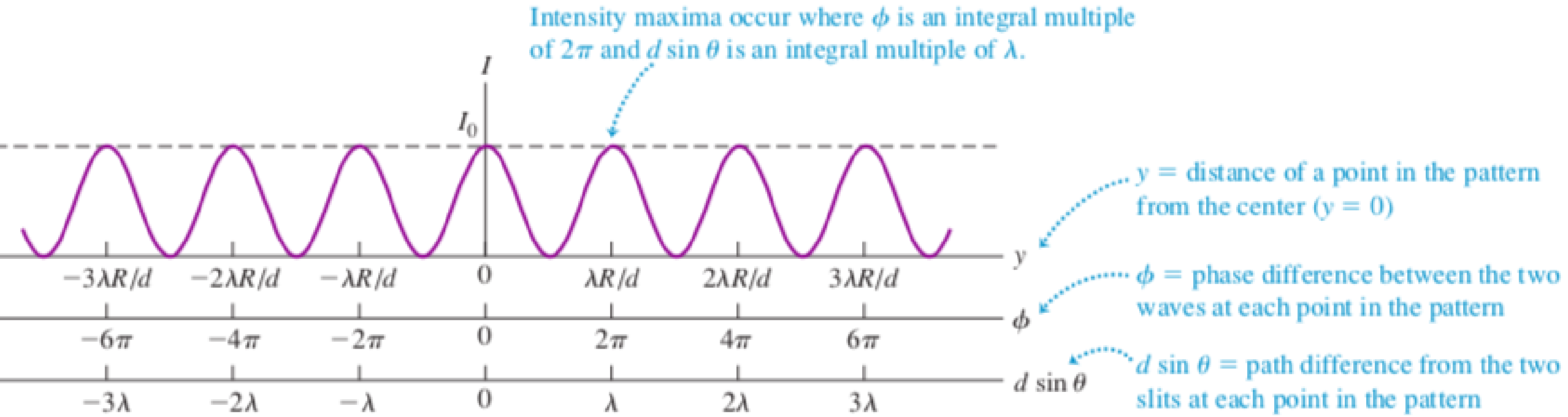
- The direction of maximum intensity:

$$\frac{\pi d\sin\theta}{\lambda} = m\pi \qquad d\sin\theta = m\lambda \qquad m = 0, \pm1, \pm2$$

# Chapter 3. Interference (3.0h)

## 3.3 Intensity in interference patterns

### Phase difference and Path difference



# Chapter 3. Interference (3.0h)

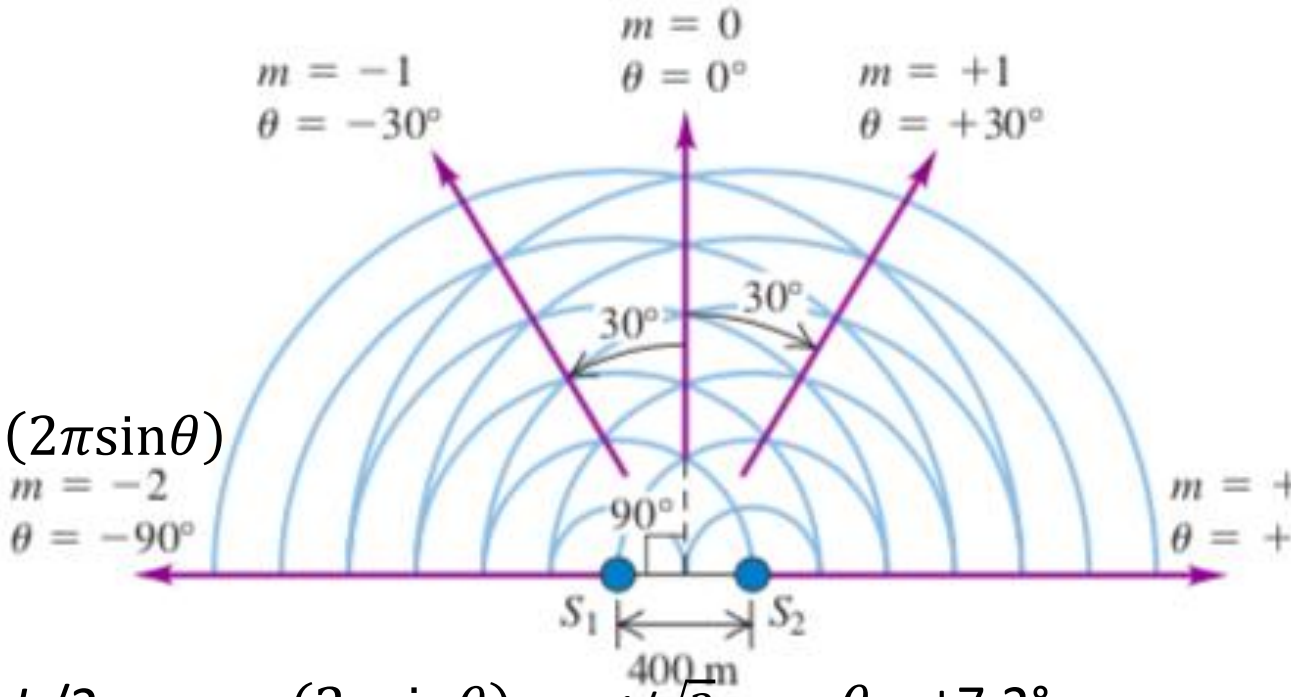
**Exp.3.3 A directional transmitting antenna array**

**Suppose the two antennas in this Figure are moved to be only 10.0 m apart and the broadcast frequency is increased to f = 60.0 MHz. At a distance of 700 m from the point midway between the antennas and in the direction $\theta$ = 0, the intensity is $I_0$ = 0.020 W/m². At this same distance, find:**

**(a) the intensity in the direction $\theta = 4.0°$;**

**(b) the direction near $\theta$ = 0 for which the intensity is $I_0/2$;**

**(c) the direction in which the intensity is zero.**




- The wavelength is

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{60 \times 10^6} = 5\ (m)$$

$$I = I_0 \cos^2\left(\frac{\pi d \sin\theta}{\lambda}\right) = I_0 \cos^2\left(\frac{10\pi \sin\theta}{5}\right) = I_0 \cos^2(2\pi \sin\theta)$$

(a) the intensity in the direction $\theta = 4.0°$;

$$I = I_0 \cos^2(2\pi \sin 4.0°) = 0.016\ \mathrm{W/m^2}$$

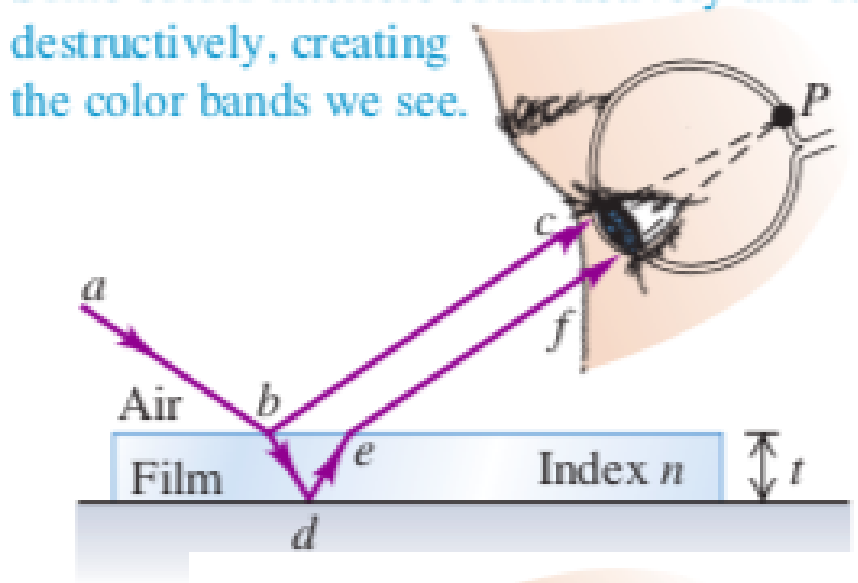
(b) the direction near $\theta$ = 0 for which the intensity is $I_0/2$; $\cos(2\pi \sin\theta) = \pm 1/\sqrt{2}$ $\theta = \pm 7.2°$

(c) the direction in which the intensity is zero. $\cos(2\pi \sin\theta) = 0$ $\theta = \pm 14.5°, \pm 48.6°$

# Chapter 3. Interferen

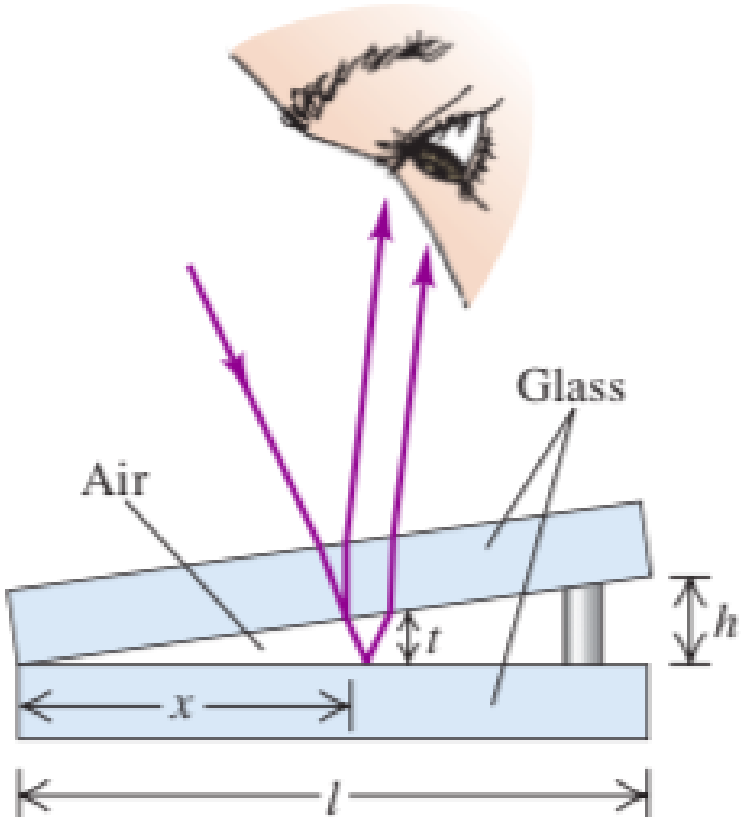
## 3.4 Interference in thin films

- Light waves are reflected from the front and back surfaces of thin films
- Constructive interference between the two reflected waves occurs: *abc* and *abdef*




### Thin-film interference and phase shifts during reflection

- Monochromatic light reflects from two nearly // surfaces at nearly normal incidence.
- Two plates of glass separated by a thin wedge of air.
- Interference between two light waves reflected from the surfaces adjacent to the air wedge.




**Constructive reflection from thin film**

$$2t = m\lambda \qquad (m = 0, 1, 2,\ldots)$$

**Destructive reflection from thin film**

$$2t = \left(m + \frac{1}{2}\right)\lambda \qquad (m = 0, 1, 2,\ldots)$$

$t$: thickness of film

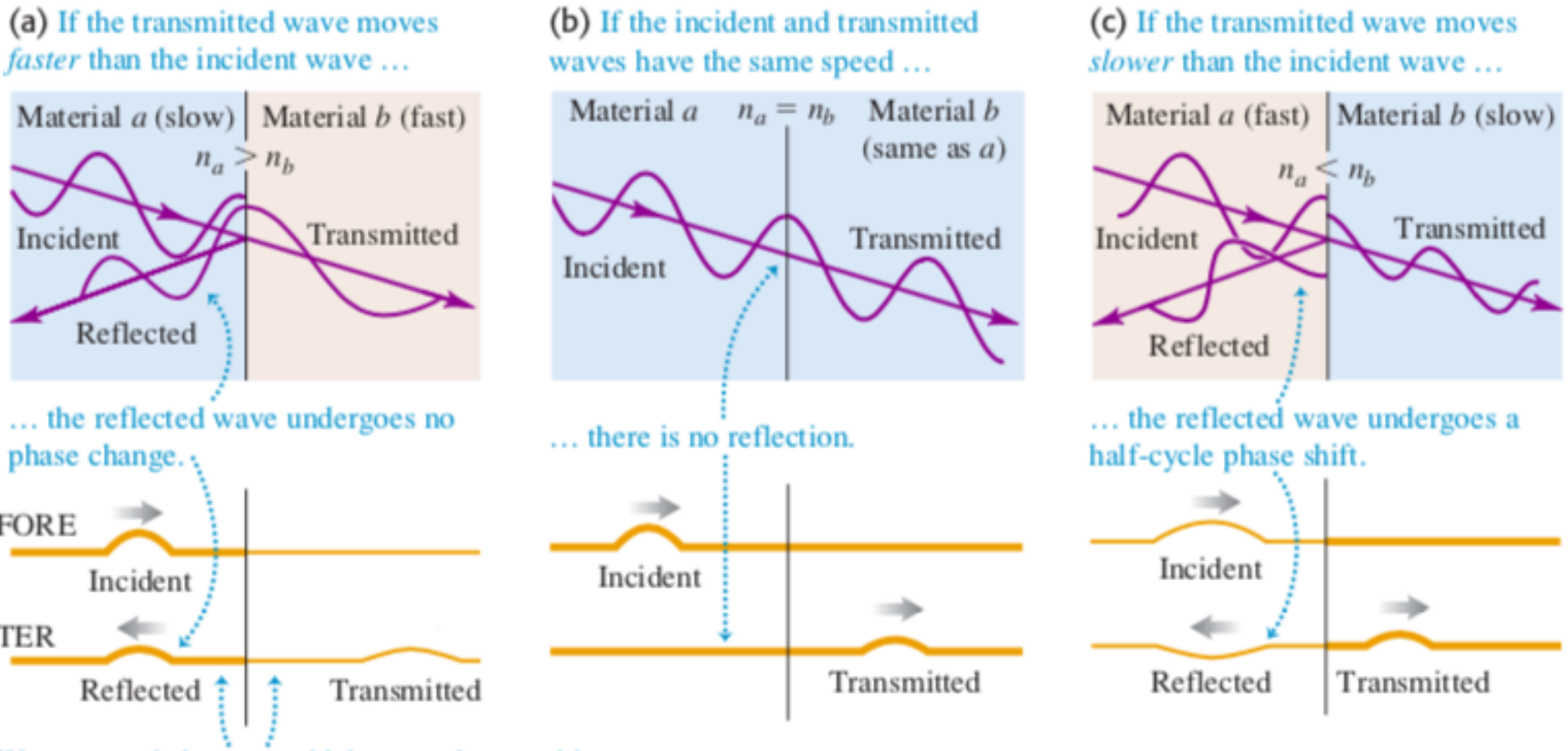
**no relative phase shift** of the reflected waves from the two surfaces

# Chapter 3. Interference (3.0h)

## 3.4 Interference in thin films

### Thin-film interference and phase shifts during reflection

**Electromagnetic waves propagating in optical materials**




**Mechanical waves propagating on ropes**




Waves travel slower on thick ropes than on thin ropes.

# Chapter 3. Interference (3.0h)

## 3.4 Interference in thin films

### Thin-film interference and phase shifts during reflection

#### Constructive reflection from thin film

$$2t = \left(m + \frac{1}{2}\right)\lambda \qquad (m = 0, 1, 2,\ldots)$$

$t$: thickness of film

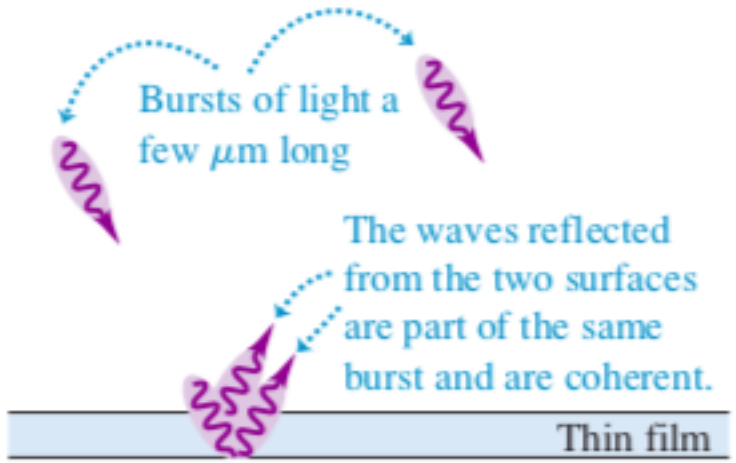
**half-cycle relative phase shift**
of one or two reflected waves
from the two surfaces

#### Destructive reflection from thin film

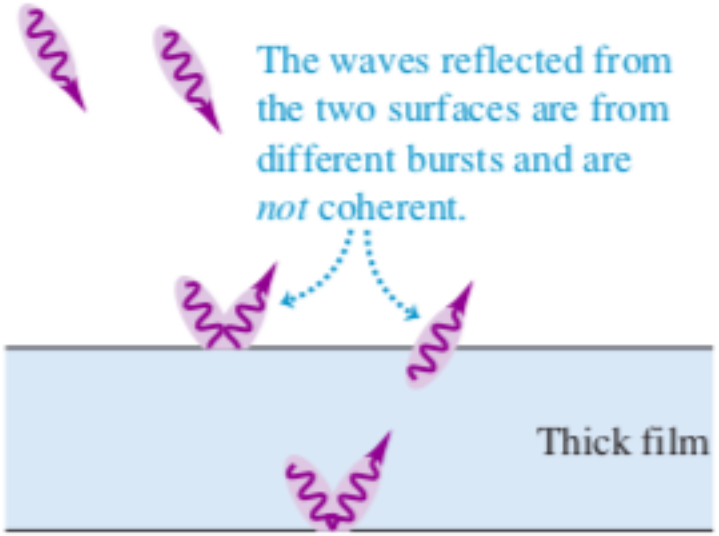
$$2t = m\lambda \qquad (m = 0, 1, 2,\ldots)$$

### Thin and thick films

(a) Light reflecting from a thin film



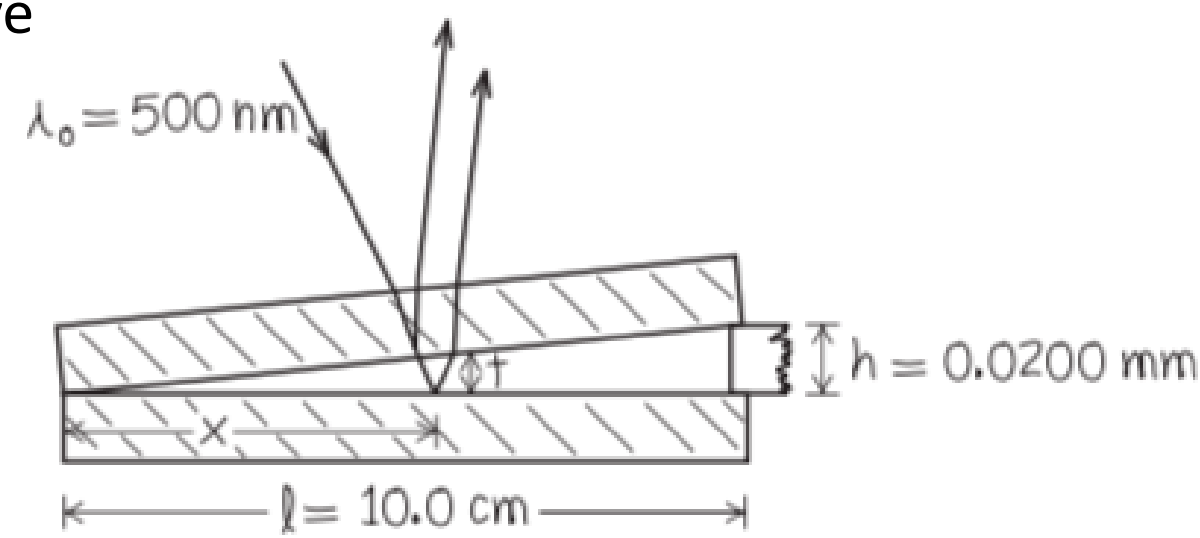

(b) Light reflecting from a thick film



**Exp.3.4 Thin-film interference I**

**Suppose the two glass plates in the figure are two microscope slides 10.0 cm long. At one end they are in contact; at the other end they are separated by a piece of paper 0.0200 mm thick. What is the spacing of the interference fringes seen by reflection? Is the fringe at the line of contact bright or dark? Assume monochromatic light with a wavelength in air of $\lambda = \lambda_0 = 500$ nm.**

- There is relative half-cycle phase shift of the wave reflected from the lower surface (air-glass).




- Destructive interference:

$2t = m\lambda$ ($m$ = 0, 1, 2,…)

$$\frac{t}{x} = \frac{h}{l}$$

$$x = m\frac{l\lambda_0}{2h} = m\frac{0.1 \times 500 \times 10^{-9}}{2 \times 0.02 \times 10^{-3}} = 1.25 \times m \text{ (mm)}$$

The spacing of the interference fringe is 1.25 mm.

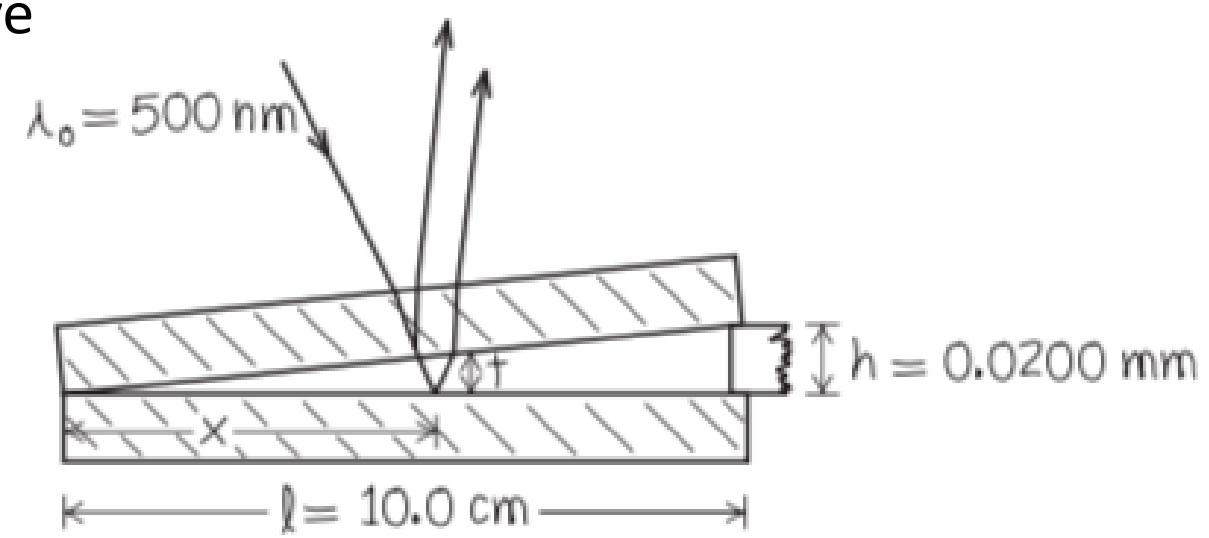
$m$ = 0 => $x$ = 0: At the line of contact, the fringe is dark.

# Chapter 3. Interference (3.0h)

**Exp.3.5 Thin-film interference II**

**Suppose the glass plates of Exp.3.4 have *n* = 1.52 and the space between plates contains water *n* = 1.33 instead of air. What happens now?**

- There is relative half-cycle phase shift of the wave reflected from the lower surface (water-glass).




- Destructive interference:

$2t = m\lambda$ ($m$ = 0, 1, 2,…)

$$\frac{t}{x} = \frac{h}{l}$$

$$x = m\frac{l\lambda_0}{n2h} = m\frac{0.1 \times 500 \times 10^{-9}}{1.33 \times 2 \times 0.02 \times 10^{-3}} = 0.94 \times m \text{ (mm)}$$

The spacing of the interference fringe is 0.94 mm.

$m$ = 0 => $x$ = 0: At the line of contact, the fringe is dark.

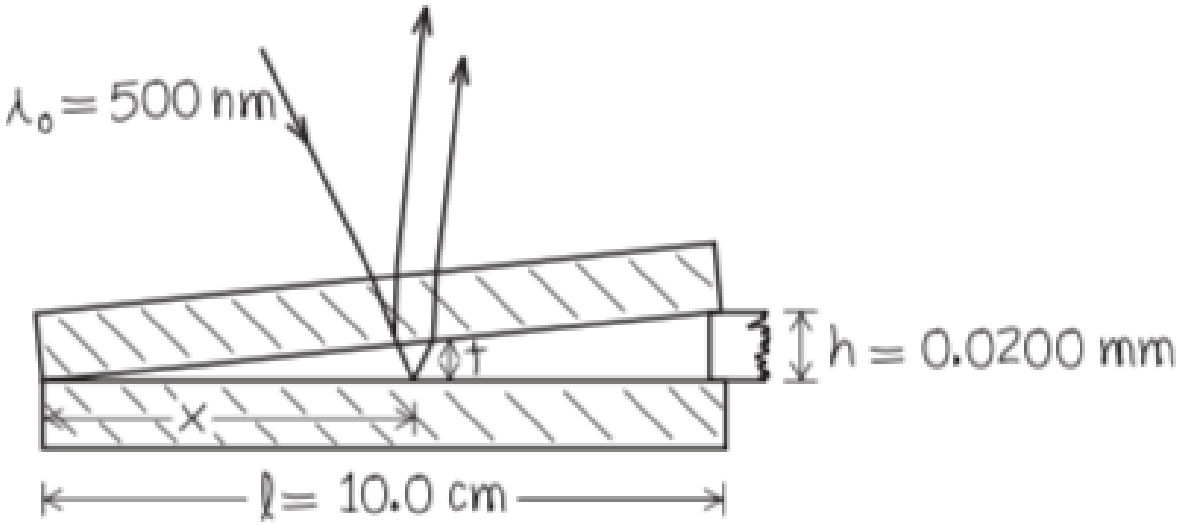
**Exp.3.6 Thin-film interference III**

**Suppose the upper of the two plates of Exp.3.4 is the plastic with *n* = 1.40, the wedge is filled with a silicone grease with *n* = 1.50, and the bottom plate is a dense flint glass with *n* = 1.60. What happens now?**

- There is no relative half-cycle phase shift of the wave reflected from the two surfaces:

+ plastic *n* = 1.40 – silicon grease *n* = 1.50: half-cycle phase shift.

+ silicon grease *n* = 1.50 – flint glass *n* = 1.60: half-cycle phase shift.




- Constructive interference:

$2t = m\lambda$ ($m$ = 0, 1, 2,…)

$$\frac{t}{x} = \frac{h}{l}$$

$$x = m\frac{l\lambda_0}{n2h} = m\frac{0.1 \times 500 \times 10^{-9}}{1.50 \times 2 \times 0.02 \times 10^{-3}} = 0.833 \times m \text{ (mm)}$$

The spacing of the interference fringe is 0.833 mm.

$m$ = 0 => $x$ = 0: At the line of contact, the fringe is bright.

# Chapter 3. Interference (3.0h)

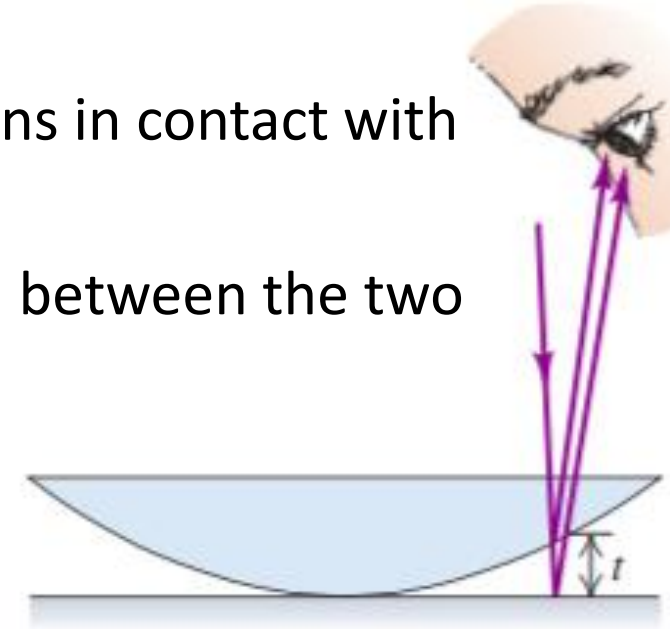
## 3.4 Interference in thin films

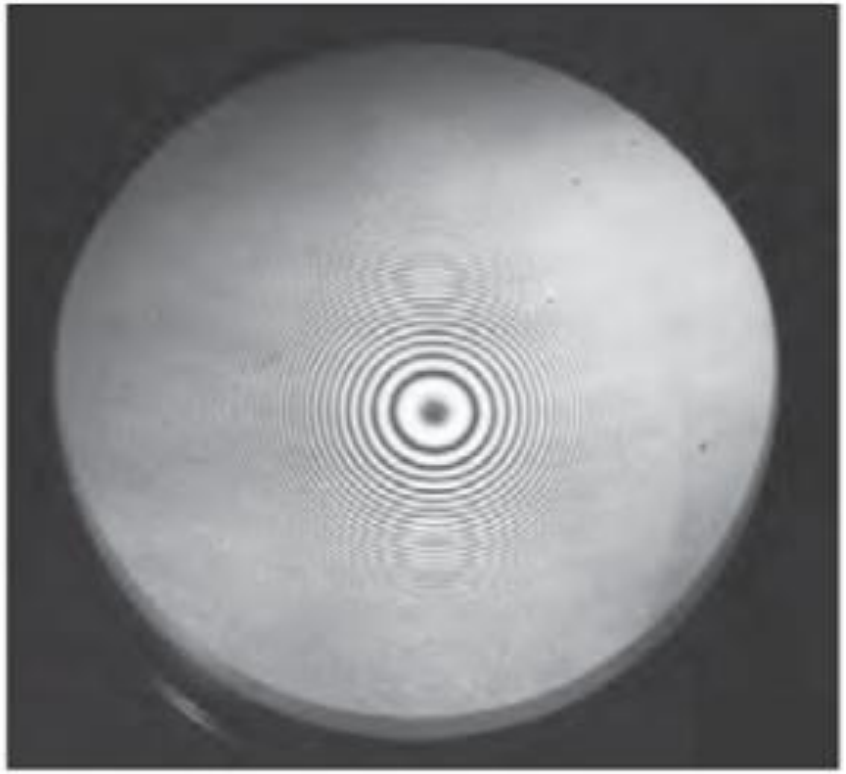
(a) A convex lens in contact with a glass plane

(b) Newton's rings: circular interference fringes

### Newton's ring

- The convex surface of a lens in contact with a plane glass plate.
- A thin film of air is formed between the two surfaces.



- View the setup with monochromatic light,

=> see circular interference fringes.

a) Air film between a convex lens and a plane surface. The thickness of the film t increases from zero as we move out from the center, giving (b) a series of alternating dark and bright rings for monochromatic light.

# Chapter 3. Interference (3.0h)

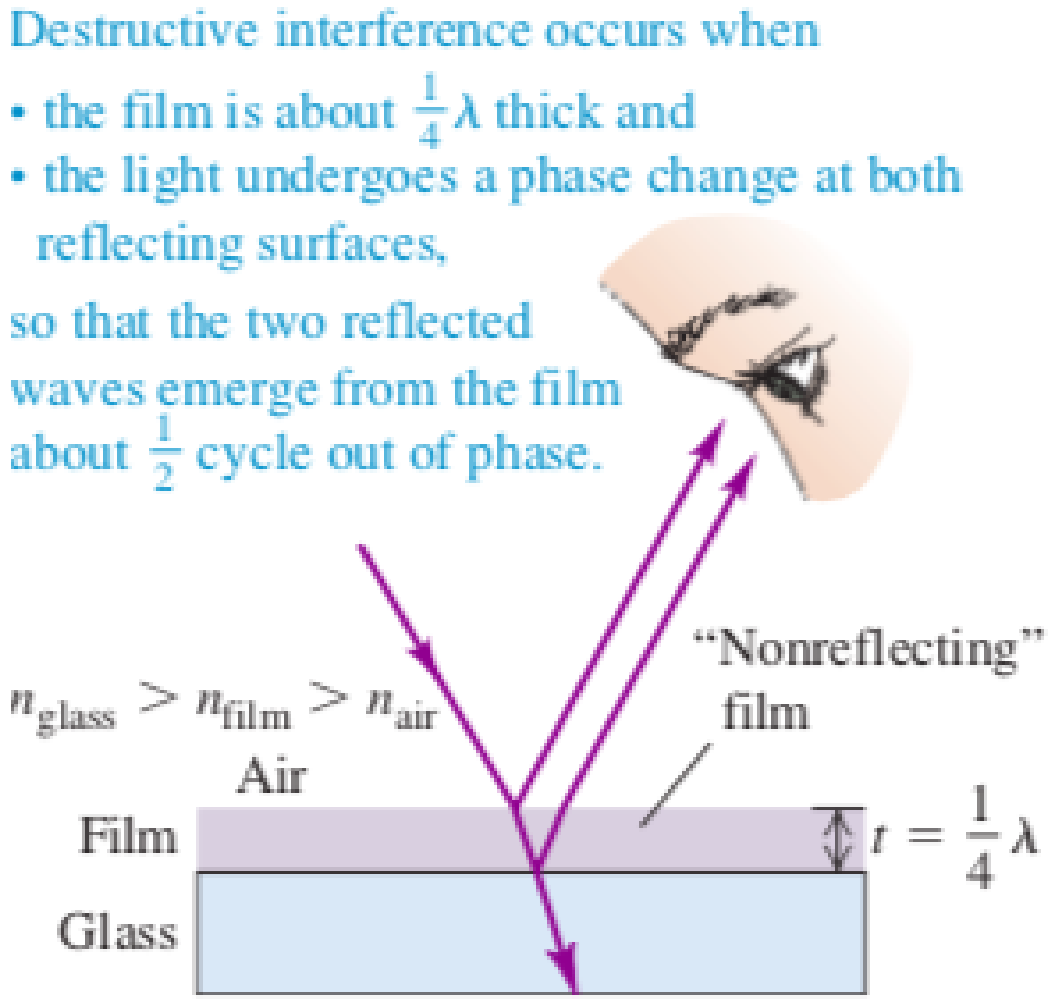
## 3.4 Interference in thin films

### Nonreflective and reflective coatings

### Nonreflective coatings

- A thin layer or film of hard transparent material with an index of refraction smaller than that of the glass is deposited on the lens surface.
- If the film thickness is a quarter (one- fourth) of the wavelength *in the film* (assuming normal incidence), the total path difference is a half-wavelength

=> destructive interference.




- The thickness of the nonreflective coating can be a quarter-wavelength for only one particular wavelength.

- If a quarter-wavelength thickness of a material with an index of refraction *greater* than that of glass is deposited on glass, then the reflectivity is *increased,* and the deposited material is called a **reflective coating.**

# Chapter 3. Interference (3.0h)

## 3.5 The Michelson interferometer

- Michelson interferometers are used to make precise measurements of wavelengths and of very small distance

$M_2$ is adjusted with a fine, very accurate micrometer screw

### How a Michelson interferometer works

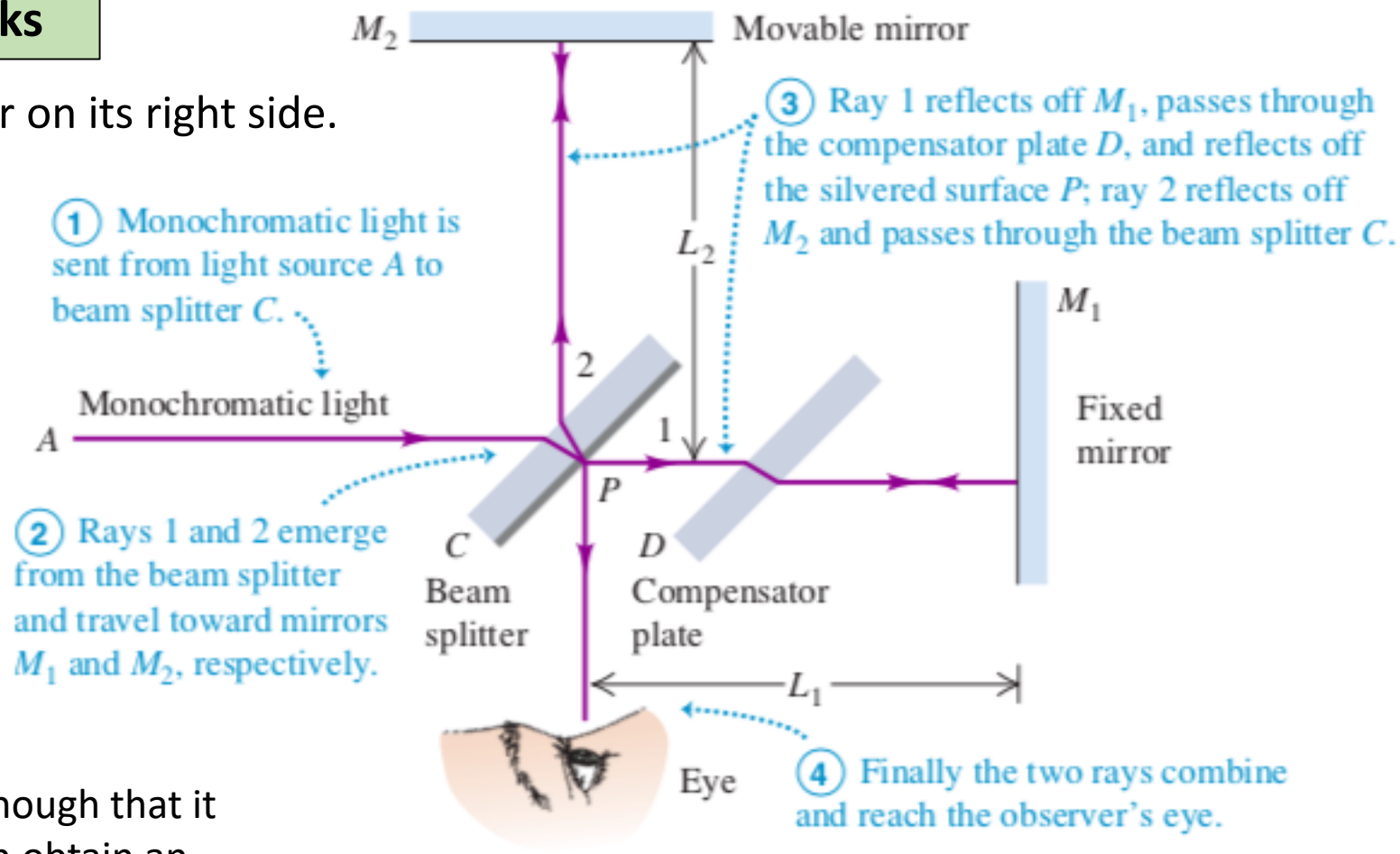
+ $C$ is a glass plate with a thin coating of silver on its right side.

+ Compensator plate $D$ is to ensure that rays 1 and 2 pass through the same thickness of glass

+ If we observe the fringe positions through a telescope with a crosshair eyepiece and $m$ fringes cross the crosshairs when we move the mirror a distance $y$, then

$$y = m\frac{\lambda}{2}; \qquad \lambda = \frac{2y}{m}$$

If $m$ is several thousand, the distance $y$ is large enough that it can be measured with good accuracy, and we can obtain an accurate value for the wavelength $\lambda$.



*“When you really want something the whole universe conspires in helping you to achieve it”*
*(The Alchemist, Paulo Coelho)*

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# GOOD LUCK and THANK YOU

*Make it Simple but Significant!*